D-FREE

MEDICATION RECOMMENDATIONS FOR RWANDA



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Organizer

Brian Lee

Jenna Shin

Organization

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Introduction

Executive Summary (Shin)

Diarrheal diseases are a leading cause of illness and death among children under five in Rwanda. These diseases significantly contribute to morbidity and mortality rates, largely due to factors such as poor sanitation, limited access to clean water, and malnutrition. This report recommends various medications as critical interventions for addressing diarrheal disease in Rwanda. Their combined use can save lives, improve recovery times, and strengthen the health resilience of vulnerable populations.

Background and Context (Shin)

Diarrheal diseases remain a significant public health challenge in Rwanda, particularly among children under the age of five. They account for approximately 10% of deaths in this age group, making them one of the leading causes of child mortality in the country. The situation is particularly severe in rural areas, where access to healthcare is limited. These regions experience higher prevalence rates, often exacerbated by factors such as malnutrition and the lack of access to safe drinking water. Several factors contribute to the widespread incidence of diarrheal diseases in Rwanda. Contaminated water sources and inadequate sanitation facilities play a substantial role in the transmission of the disease. Environmental challenges, coupled with high poverty rates and limited access to healthcare services, further compound the issue. Additionally, the lack of widespread public health education leaves many communities unaware of preventive measures. Malnutrition also significantly exacerbates the problem, as weakened immune systems leave children more vulnerable to severe cases of diarrhea. In response to this critical public health issue, both the Rwandan government and various non-governmental organizations (NGOs) have implemented a range of interventions. Efforts have been made to promote the use of improved sanitation systems and safe drinking water. Initiatives to expand the availability of oral rehydration therapy (ORT), zinc supplements, vitamin A, and probiotics supplementation have been introduced, although significant gaps persist, particularly in rural areas. Public health campaigns focused on hygiene practices and disease prevention have also been launched, aiming to educate communities and reduce the incidence of diarrheal diseases. Despite these efforts, challenges remain, underscoring the need for sustained and targeted interventions.

Learning Objectives (Lee)

By the end of this report, medical practitioners should be able to:

- Acknowledge the significance of the suggested medications as effective solutions for diarrheal diseases.
- Properly prepare the suggested medications for corresponding age groups and severity of the disease.
- Ensure accurate storage and management of the suggested medications for their effectiveness.
- Acknowledge the limitations of each medication and effectively respond to signs of potential side-effects.

Recommended Medication

Overview (Lee)

Treatment	Formulation	Mechanism of Action
Oral Rehydration Therapy (ORT)	Low-osmolarity Oral Rehydration Solution (ORS) in powder or liquid form.	Replaces lost fluids and electrolytes, preventing life-threatening dehydration caused by diarrhea.
Zinc Supplements	Tablets (20 mg) or syrup.	Boosts the immune system, reduces the severity of diarrhea, and prevents recurrence.
Vitamin A Supplements	High-dose liquid drops and capsules.	Stimulate production of antibodies and supports the function of epithelial cells for stronger gut integrity.
Probiotic Supplements	Capsules with specific probiotic strains*.	Creates a healthier gut microbiome and supports the function of epithelial cells for stronger gut integrity.

Rationale for medication

Oral Rehydration Therapy (ORT) (Shin)

Oral Rehydration Therapy (ORT) is a highly effective treatment for diarrheal diseases and has been extensively proven to reduce stool output and hospitalization rates. Decades of evidence highlight its success, particularly in resource-limited settings, making it an ideal solution for addressing the burden of diarrheal diseases in Rwanda. Its safety profile is excellent, with minimal side effects, making it suitable for children and adults alike. Furthermore, ORS packets are stable at room temperature, eliminating the need for specialized storage facilities and ensuring their practicality in rural areas. The affordability of ORT, costing less than $0.50 per treatment course, makes it accessible to even the most economically disadvantaged families. Moreover, its compatibility with Rwanda's existing healthcare infrastructure allows for easy integration into community health programs and clinics, ensuring efficient distribution and administration in low-resource settings.

Zinc Supplements (Shin)

Zinc supplementation is another critical component in the treatment and prevention of diarrheal diseases. Clinical studies have shown that zinc can reduce the duration of diarrhea by 25% and decrease the likelihood of recurrence by 40%. This is particularly beneficial for malnourished children, who are more vulnerable to severe diarrhea and its complications. Like ORT, zinc supplements have a strong safety profile, with minimal side effects, and do not require any special storage conditions. They are both practical and sustainable, even in remote areas. The cost-effectiveness of zinc is also notable, with each tablet priced at approximately $0.02 to $0.05, making it feasible for widespread use. Given its ease of administration, zinc supplements can seamlessly complement Rwanda's healthcare infrastructure, especially through community health workers and local clinics.

Vitamin A Supplementation (Lee)

Vitamin A supplementation is a crucial medication that could provide support for the residents in Rwanda who are struggling with diarrheal diseases. Vitamin A is a commonly known nutrient to have various benefits to one's body system. Vitamin A is excellent in forming stronger immune system, supporting the function of white blood cells to combat against harmful pathogens and bacteria. Furthermore, the nutrient greatly promotes the integrity of mucosal barriers and is a crucial factor for cell growth and differentiation. It can support the growth and function of epithelial cells: cells that form epithelial barriers, also known as a layer of the gut. Hence, one with vitamin A supplementation is capable of maintaining a strong intestine, effectively reducing the impact of diarrheal diseases. Although the price of high-dose tablets could vary from $0.10 to $0.30 per capsule, they are readily accessible through various means of purchase. Ultimately, Vitamin A is an effective intervention for public health programs that could significantly help the Rwandan community, playing a pivotal role in alleviating the burden of the health issue they are currently facing.

Probiotic Supplementation (Lee)

Probiotic supplementation could also be a key intervention to mitigate diarrheal diseases in Rwanda. Studies of probiotics have been an emerging topic in the medical and nutritional field for the past years, as their significance in promoting gut health and supporting the immune system has become increasingly evident. Supplementation of probiotics, specifically, lactobacillus rhamnosus and saccharomyces boulardii*, promotes the abundance of beneficial gut bacteria in one's intestine to create a healthy gut microbiome, as well as supporting the growth and function of epithelial cells for the maintenance of a strong gut layer. Both benefits are crucial for a stronger resistance against gut-related diseases. Proven for their excellence to fulfill such goals, probiotic supplementation could likely to be one of the most effective intervention for public health program in Rwanda, making the population capable of strongly combatting against diarrheal diseases.

Medication Guidance

Dosage & Administration (Lee)

Oral Hydration Therapy

Standard ORT Packet Composition (per liter of water): 2.6 g sodium chloride, 13.5 g glucose (anhydrous), 1.5 g potassium chloride, 2.9 g trisodium citrate dihydrate.

Dissolve one ORT packet in 1 L of clean, portable water. Administer after each loose stool. Children under 5 should administer 100 - 200 mL per dosage, 10 - 20 mL per body weight (kg). Children over 5 and adults should follow the same preparation; 200 - 400 mL per dosage. Dosage adjusted for body weight and severity.

Zinc Supplement

Prepare elemental zinc in capsule or syrup form.

Children under 6 months should intake 10 mg daily of elemental zinc in capsule or syrup form for 10 - 14 days, depending on severity. Children between the range of 6 months and 5 years old should intake 20 mg daily for the same duration. Children over 5 and adults should intake 20 - 40 mg daily for the same duration. Intake food with dosage.

Vitamin A Supplementation

Prepare capsules of vitamin A with corresponding IU. Cut and squeeze liquid into patient's mouth.

Children under 6 months should receive a single oral dose with 50,000 IU. Children between the range of 6 and 11 months should receive a single oral does with 100,000 IU. Children over 12 months should receive the same dose with 200,000 IU. Adults equivalent children over 12 months.

Probiotic Supplement

Prepare capsules of probiotics (lactobacillus rhamnosus, saccharomyces boulardii) with corresponding CFU. Intake capsules with water.

Children under 5 years old should intake probiotic capsules of 10^9 CFU twice a day. Children over 5 and adults should intake the same with 10^10 CFU. Both age groups should continue dosage for 7 consecutive days.

Storage

Side Effect & Solution (Lee)

Oral Health Therapy

Storage

ORT packets should be stored in cool, dry areas (15°C - 30°C) with avoidance from sunlight or moisture. ORT packet mixed with water should be stored in the same environment and used within 24 hours.

Side Effects and solutions

- Vomiting due to rapid intake.

Wait for 10 to 15 minutes until the patient vomits and recovers. Provide small, frequent sips of ORT mixture every 1 to 2 minutes.

- Stomach discomfort due to electrolyte imbalance and rapid intake.

Correct mixture of ratio between ORT and water is essential. Provide small, frequent sips of ORT mixture every 1 to 2 minutes once the patient is comfortable.

- Lethargy, confusion (rare).

Seek immediate support from medical experts.

Zinc Supplement

Storage

Zinc capsules should be stored in cool, dry areas (15°C - 30°C) with avoidance from sunlight or moisture. Zinc liquid should be sealed in a closed container under the same environement.

Side Effects and solutions

- Stomach irritation due to intake with an empty stomach.

Zinc supplement should be done with food intake, as well as with plenty of water.

- Stomach cramp, headache or fatigue due to high doses and overuse.

The patient should receive an accurate amount of zinc supplement without excess. Continue intake once patient recovers and feels comfortable.

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Vitamin A Supplement

Storage

Vitamin A drops and capsules should be stored in cool, dry areas (15°C - 35°C) with avoidance from sunlight or moisture.

Side Effects and solutions

- Vomiting, headache, or dizziness due to high doses and overuse.

The patient should receive the an accurate amount of daily vitamin A supplements. Continue intake once patient feels comfortable. Stop intake if the patient shows constant discomfort.

- Ineffectiveness due to incomplete intake.

Make sure the patient swallows vitamin A drops completely. Monitor situation especially for infants. Avoid intake with hot water, due to potential vitamin A degradation.

Probiotic Supplement

Storage

Avoidance from sunlight and moisture is essential. supplement with lactobacillus rhamnosus should be kept under refrigeration (2°C - 8°C). Saccharomyces boulardii should be kept under room temperature. Capsule label should be checked before storage.

Side Effects and solutions

- Temporal gas, bloating due to adjustment of gut flora.

Continue dosage after patient feels comfortable. Start with a low dosage and increase gradually.

- Mild diarrhea due to temporal effects on gut.

Reduce dosage temporally. Ensure hydration and intake probiotic supplements with food.

Stock Management & Supply Chain Considerations

(Lee)

Oral Rehydration Therapy (ORS)

- Training workers to ensure and track accurate storage mechanism for long-term usage.
- Manage sufficient supply of ORS in healthcare facility.
- Attainable through international organizations (UNICEF, WHO, etc).

Zinc Supplement

- Training workers to ensure and track accurate storage mechanism for long-term usage.
- Manage sufficient supply of zinc supplement in healthcare facility.
- Directly purchasable through local pharmacy / importation.

Vitamin A Supplement

- Training workers to ensure and track accurate storage mechanism for long-term usage.
- Manage sufficient supply of vitamin A in healthcare facility.
- Attainable through both international organizations (UNICEF, WHO, etc) and direct purchase.

Probiotic Supplement

- Training workers to ensure and track accurate storage mechanism for long-term usage.
- Manage sufficient supply of probiotics in healthcare facility.
- Directly purchasable through local pharmacy / importation.

Conclusion

End of Report (Lee)

The overall use of Oral Rehydration Therapy (ORT), zinc, vitamin A, and probiotic supplements offer a life-saving, as well as accessible solution to significantly resist against diarrheal diseases in Rwanda. By strengthening the healthcare system's capacity to address these diseases and promoting community awareness, these medications can significantly mitigate the health-related burden the targeted community is currently facing. **Let's work together to promote the proper and effective use of the medications to support health facilities across Rwanda.** For further questions, please refer to national health guidelines or visit the D-Free website for inquiries.

Reference

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